

2026 FALL SEMESTER
Programming for Data Science (MACSE502) Lab Record
Practice Program

Date: 23/07/2026

Week No.: 3

Program S. No. 5

Title	Creating and Manipulating Arrays Using NumPy				
Problem Statement	Create a 2D array of shape (3, 4) with random integers between 0 and 10. Perform the following operations: calculate the mean and standard deviation of the entire array, slice the array to get the first two rows and last two columns, reshape the array to shape (4, 3), and perform element-wise multiplication with another array of the same shape				
Objectives	• To create and manipulate two-dimensional NumPy arrays. • To represent predefined array values using a Python dictionary. • To calculate the mean and standard deviation of the entire array. • To slice the array and extract selected rows and columns. • To reshape an array from shape (3, 4) to shape (4, 3). • To perform element-wise multiplication between arrays of the same shape. • To understand array shape, slicing, reshaping and numerical operations using NumPy.				
Dataset URL Link / File	Predefined dictionary dataset used directly inside the Python program:				
	Array	Row 1	Row 2	Row 3	Row 4
	Array1 (3 x 4)	2, 5, 7, 1	8, 3, 6, 4	9, 0, 5, 2	—
	Array2 (4 x 3)	1, 2, 3	4, 5, 6	7, 8, 9	2, 3, 4
	No external dataset file or URL is required.				
Functions Details	Sl. No.	Function	Package	Description of function	
	1	np.array()	NumPy	Converts dictionary values into NumPy arrays.	
	2	shape	NumPy	Returns the dimensions of an array as rows and columns.	
	3	np.mean()	NumPy	Calculates the mean of all elements in the array.	
	4	np.std()	NumPy	Calculates the standard deviation of all array elements.	
	5	Array slicing	NumPy	Extracts the first two rows and last two columns.	
	6	reshape()	NumPy	Changes the array shape from (3, 4) to (4, 3).	
	7	round()	Python	Rounds numerical output to the specified decimal places.	
	8	print()	Python	Displays arrays, calculations and headings in the console.	
	9	range()	Python	Creates a sequence of integer values when required.	
	10	len()	Python	Returns the number of items in an object.	
	11	dtype	NumPy	Shows the data type of array elements.	

Student Reg. No: 26MML0031 Name: Makada Hasnain

Faculty: Dr Rajasekhara Babu M, Professor, School of CSE, Vellore Institute of Technology (VIT), Vellore

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	12	Element-wise *	NumPy	Multiplies corresponding elements of equal-shaped arrays.
Steps/ Process/ Procedure/ Algorithm/ Flowchart/ Model	Step 1: Import NumPy using import numpy as np. Step 2: Create a predefined dictionary containing Array1 and Array2. Step 3: Convert the dictionary values into NumPy arrays using np.array(). Step 4: Display Array1 and verify its shape as (3, 4). Step 5: Calculate the mean using np.mean(). Step 6: Calculate the standard deviation using np.std(). Step 7: Slice Array1 using array1[:2, -2:] to get the first two rows and last two columns. Step 8: Reshape Array1 to (4, 3) using reshape(4, 3). Step 9: Display Array2 and verify its shape as (4, 3). Step 10: Perform element-wise multiplication using reshaped_array * array2. Step 11: Display all results and their respective shapes. Step 12: Verify that all requested NumPy operations are completed successfully.			
Program	<pre>import numpy as np array_data = { "Array1": [[2, 5, 7, 1], [8, 3, 6, 4], [9, 0, 5, 2]], "Array2": [[1, 2, 3], [4, 5, 6], [7, 8, 9], [2, 3, 4]] } array1 = np.array(array_data["Array1"]) array2 = np.array(array_data["Array2"]) print("ORIGINAL 2D ARRAY") print(array1) print("Shape:", array1.shape) mean_value = np.mean(array1) standard_deviation = np.std(array1) print("\nMean:", round(mean_value, 2)) print("Standard Deviation:", round(standard_deviation, 2)) sliced_array = array1[:2, -2:] print("\nFIRST TWO ROWS AND LAST TWO COLUMNS") print(sliced_array) print("Shape:", sliced_array.shape) reshaped_array = array1.reshape(4, 3) print("\nRESHAPED ARRAY")</pre>			

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```
print(reshaped_array)
print("Shape:", reshaped_array.shape)

print("\nSECOND ARRAY")
print(array2)

multiplied_array = reshaped_array * array2
print("\nELEMENT-WISE MULTIPLICATION")
print(multiplied_array)
print("Shape:", multiplied_array.shape)

print("\nFINAL SUMMARY")
print("Original Shape:", array1.shape)
print("Mean:", round(mean_value, 2))
print("Standard Deviation:", round(standard_deviation, 2))
print("Sliced Shape:", sliced_array.shape)
print("Reshaped Shape:", reshaped_array.shape)
print("Multiplied Shape:", multiplied_array.shape)
```

**Output
Snapshots
With
Explanation/
Observations**

Screenshot 1: Original Array and Shape

ORIGINAL 2D ARRAY

```
=====
[[2 5 7 1]
 [8 3 6 4]
 [9 0 5 2]]
```

Shape of Array: (3, 4)

Figure 1.1 Original 2D Array

The predefined dictionary values were converted into a NumPy array. The output confirms that the array contains 3 rows and 4 columns.

Screenshot 2: Mean and Standard Deviation

MEAN OF THE ENTIRE ARRAY

```
=====
Mean: 4.33
```

STANDARD DEVIATION OF THE ENTIRE ARRAY

```
=====
Standard Deviation: 2.72
```

Figure 1.2 Mean & Standard Deviation of the Array

The mean represents the average of all elements in the array, while the standard deviation shows how widely the values are distributed around the mean.

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Screenshot 3: Array Slicing

```
FIRST TWO ROWS AND LAST TWO COLUMNS
```

```
=====
[[7 1]
 [6 4]]
```

Figure 1.3 Sliced Array

Array slicing was performed using `array1[:2, -2:]`. The resulting sliced array has a shape of 2×2 .

Screenshot 4: Reshaped Array

```
RESHAPED ARRAY
```

```
=====
[[2 5 7]
 [1 8 3]
 [6 4 9]
 [0 5 2]]
```

```
Shape After Reshaping: (4, 3)
```

Figure 1.4 Reshaped Array

The `reshape()` function changed the arrangement of the 12 elements without changing their values or total count.

Screenshot 5: Element-wise Multiplication

```
ELEMENT-WISE MULTIPLICATION
```

```
=====
[[ 2 10 21]
 [ 4 40 18]
 [42 32 81]
 [ 0 15  8]]
```

Figure 1.5 Element Wise Multiplication

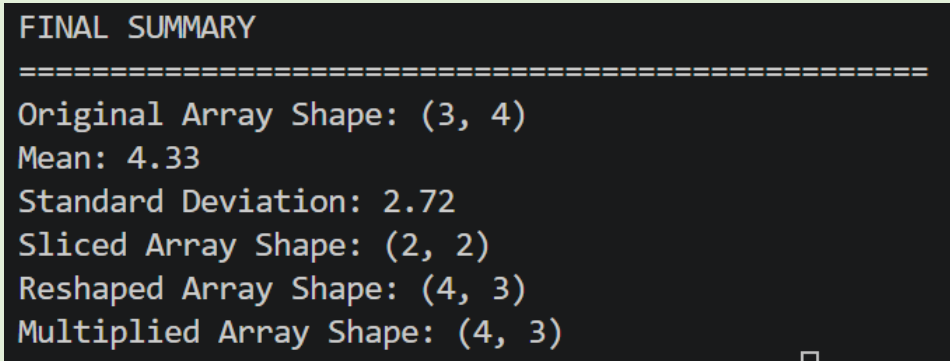
Corresponding elements of the two arrays with shape (4, 3) were multiplied using the `*` operator.

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	Screenshot 6: Final Summary  <i>Figure 1.6 Final Summary</i> The final output verifies that all statistical, slicing, reshaping and multiplication operations were completed successfully.
Conclusions	The NumPy array manipulation exercise was completed successfully. A two-dimensional array of shape (3, 4) was created using predefined dictionary data. The mean and standard deviation were calculated, the first two rows and last two columns were extracted through slicing, and the array was reshaped to (4, 3). Element-wise multiplication was performed with another array of the same shape. The exercise provided practical understanding of NumPy array creation, statistical calculations, slicing, reshaping and element-wise operations.
References	1. Michael Freeman and Joel Ross, Programming Skills for Data Science: Start Writing Code to Wrangle, Analyze, and Visualize Data with R, Addison-Wesley, 2018. 2. NumPy documentation for array creation, statistics, slicing, reshaping and mathematical operations. 3. Python program link (https://github.com/hasnainmakada-99/Data-Science-Codes/blob/main/WEEK%203/PRACTICE_PROGRAM.py) 4. Programming for Data Science (MACSE502) class lecture notes.